

Class 8 Mini-Project: Unsupervised Learning Analysis of Human Breast Cancer Cells

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Background

The goal of this mini-project is for us to explore a complete analysis using unsupervised learning techniques covered in class.

Today we will analyze a data-set biopsy from fine needle aspiration (FNA) of a breast mass.

Data Import

The data is made available as a CSV file for download. We can read this in using `read.csv()`

```
wisc.df <- read.csv("WisconsinCancer.csv", row.names=1)
```

```
head(wisc.df, 4)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302	0.11840	0.27760	0.3001		0.14710
842517	0.08474	0.07864	0.0869		0.07017
84300903	0.10960	0.15990	0.1974		0.12790
84348301	0.14250	0.28390	0.2414		0.10520
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419	0.07871	1.0950	0.9053	8.589
842517	0.1812	0.05667	0.5435	0.7339	3.398
84300903	0.2069	0.05999	0.7456	0.7869	4.585
84348301	0.2597	0.09744	0.4956	1.1560	3.445
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003	0.006193	25.38	17.33	
842517	0.01389	0.003532	24.99	23.41	
84300903	0.02250	0.004571	23.57	25.53	
84348301	0.05963	0.009208	14.91	26.50	
	perimeter_worst	area_worst	smoothness_worst	compactness_worst	
842302	184.60	2019.0	0.1622	0.6656	
842517	158.80	1956.0	0.1238	0.1866	
84300903	152.50	1709.0	0.1444	0.4245	
84348301	98.87	567.7	0.2098	0.8663	
	concavity_worst	concave.points_worst	symmetry_worst		
842302	0.7119	0.2654	0.4601		
842517	0.2416	0.1860	0.2750		
84300903	0.4504	0.2430	0.3613		
84348301	0.6869	0.2575	0.6638		
	fractal_dimension_worst				
842302	0.11890				
842517	0.08902				
84300903	0.08758				
84348301	0.17300				

We have to make sure to remove/exclude the `diagnosis` column from the data-set as it is the

expert diagnosis as either M or B. To remove the first column we can create a new data.frame that omits it.

```
wisc.data <- wisc.df[,-1]  
diagnosis <- as.factor( wisc.df$diagnosis )
```

Exploratory data analysis

Q1. How many observations are in this dataset?

```
nrow(wisc.data)
```

```
[1] 569
```

There are 569 observations in this data set

Q2. How many of the observations have a malignant diagnosis?

```
table(wisc.df$diagnosis)
```

```
  B   M  
357 212
```

212 of the observations have a malignant diagnosis

Q3. How many variables/features in the data are suffixed with `_mean`?

We can use the `grep()` function to help us here:

```
length(grep("_mean", colnames(wisc.data), value = TRUE))
```

```
[1] 10
```

10 variables/features in the data are suffixed with `_mean`

Principal Component Analysis (PCA)

Before performing PCA we must scale our data with the `scale = TRUE` argument to `prcomp` because: - The input variables use different units of measurement - The input variables have significantly different variances

```
wisc.pr <- prcomp(wisc.data, scale = TRUE)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010

	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335

	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966

	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997

	PC29	PC30
Standard deviation	0.02736	0.01153
Proportion of Variance	0.00002	0.00000
Cumulative Proportion	1.00000	1.00000

Q4. From your results, what proportion of the original variance is captured by the first PC1?

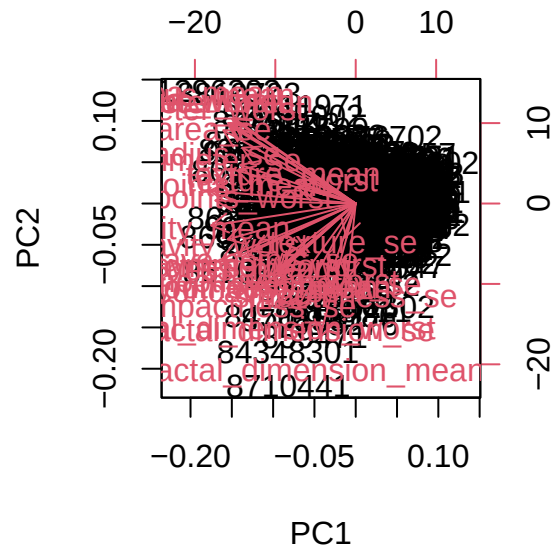
From the results the proportion of the original variance that is captured by the first PC1 is 0.4427.

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

3 principal components are required to describe at least 70% of the original variance in the data.

Biplot of the `wisc.pr`

```
biplot(wisc.pr)
```



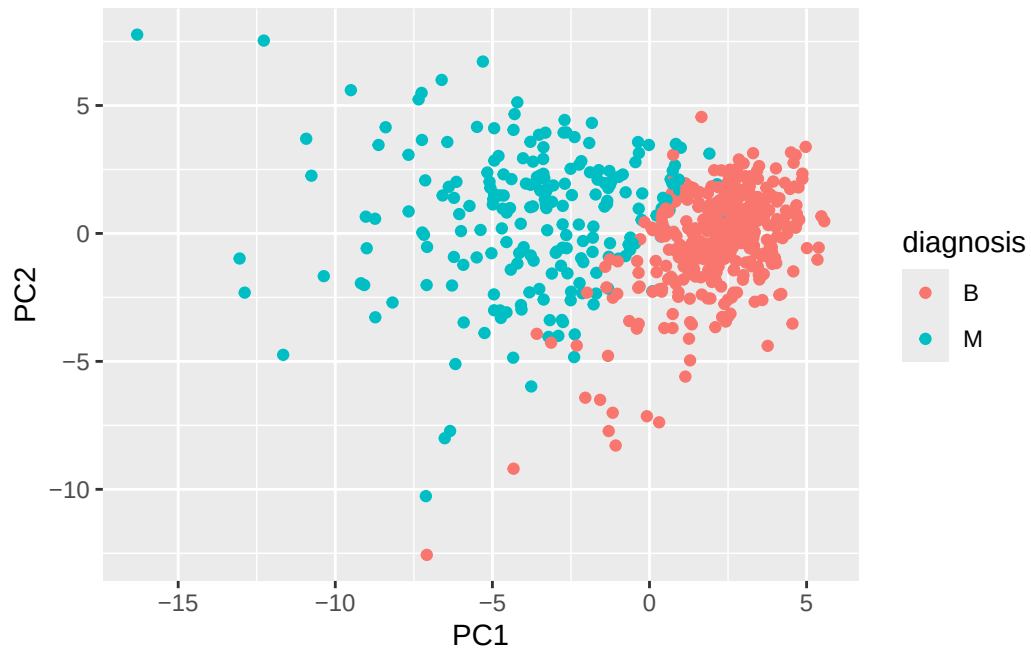
Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

One thing that stands out to me about this plot is the amount of information that seems to be clustered in it making it rather difficult to understand given that you can't read much of the text or numbers in the graph.

Let's see the "PC score plot"

```
library(ggplot2)

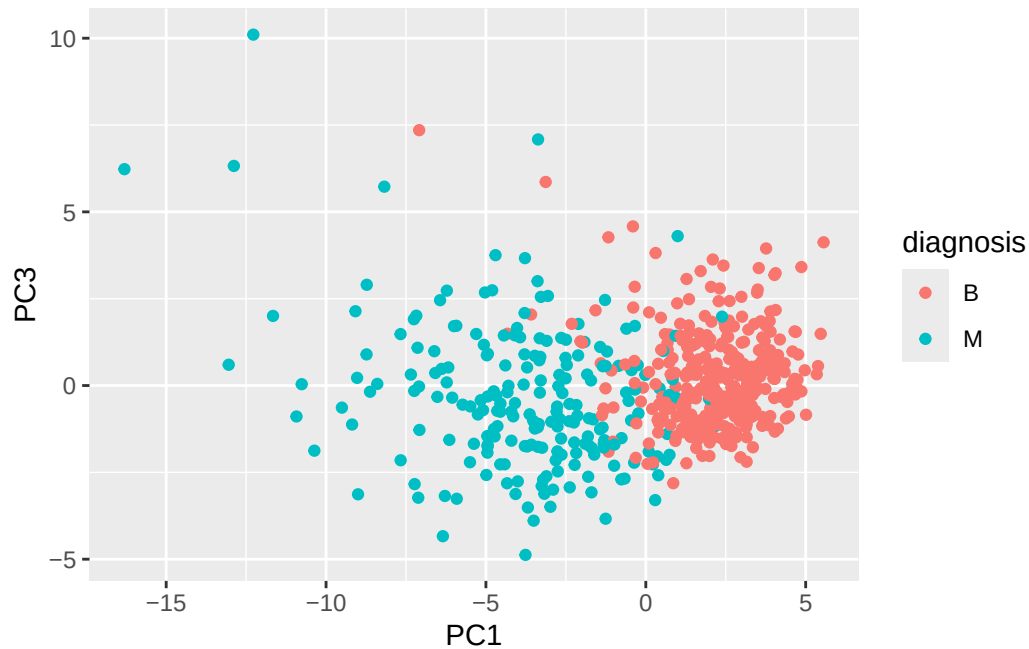
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col = diagnosis) +
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

Both plots indicate a separation between malignant and benign samples through PC1.

```
ggplot(wisc.pr$x) +  
  aes(PC1,PC3, col=diagnosis) +  
  geom_point()
```



The loading vector `wisc.pr$rotation` tells us which original measurements contribute most to each PC.

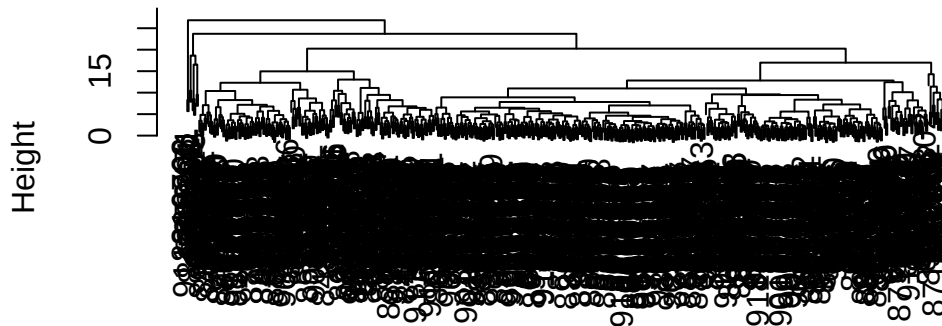
Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`?

For the first principal component, the component of the loading vector for the feature `concave.points_mean` is -0.26085376

Hierarchical clustering

```
wisc.hclust <- hclust(dist(scale(wisc.data)))
plot(wisc.hclust)
```

Cluster Dendrogram



```
dist(scale(wisc.data))  
hclust (*, "complete")
```

Combining Methods

```
d <- dist(wisc.pr$x[,1:4])  
wisc.pr.hclust <- hclust(d, method = "ward.D2")  
plot(wisc.hclust)  
abline(h=70, col = "red")
```


Cluster Dendrogram



```
dist(scale(wisc.data))  
hclust(*, "complete")
```

```
grps <- cutree(wisc.pr.hclust, h=70)  
table(grps)
```

```
grps  
  1   2  
171 398
```

How does this clustering `grps` correspond to the expert diagnosis?

```
table(diagnosis, grps)
```

```
      grps  
diagnosis  1   2  
  B      6 351  
  M     165  47
```

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses?

The hierarchical clustering models created in the previous sections in which we didn't first do PCA was not very good in terms of separating the diagnoses.

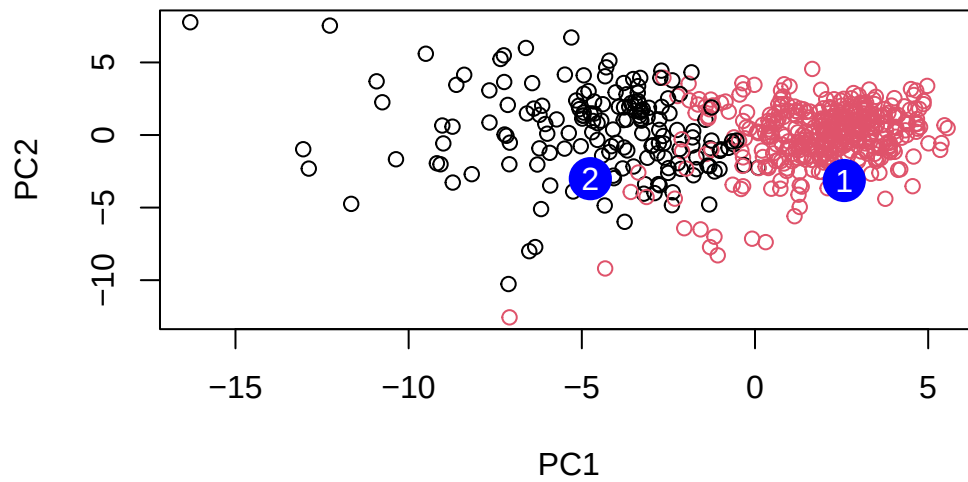
Prediction

We will use the `predict()` function that will take our PCA model from before and new cancer cell data and project that data onto our PCA space.

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Based on these results we should prioritize patients in section 2 as they have more malignant diagnosis compared to patients from section 1.